

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-I

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Dr.B.Deepa Lakshmi

Name of the Topic: Control System Basic Structure, Block Diagram Models - Reduction and Signal Flow Graphs

Name of the Innovative Practice: Zero Minute Speech, Collaborative Study and Minute Paper

Date & Duration: 01.07.2019 (10 Minutes), 08.07.2019 (20 Minutes) & 10.07.2019 (10 Minutes)

Description:

(i) Zero Minute Speech:

Zero Minute Speech enables the students to explain the topic in short in a better way. This can be understood by other students easily. So, the students will listen the speech and they will be ready to give speech on the topic.

(ii) Collaborative Study:

Collaborative learning is used to learn about any topic within a group. When compared to self learning, students are engaged in groups to discuss and learn among them.

(iii) Minute Paper:

It enables the teacher to measure the level of understanding the topic by the students. It enhances the thinking ability of the students. The student will remember the concept once they write it immediately and remember for a longer period of time.

Goals (Learning Outcomes):

1. The students will be able to explain about the basic components of control system.
2. The students will be able to reduce the block diagram and determine transfer function.
3. The students will be able to obtain the transfer function of the given system using signal flow graph models.

Use of appropriate method:

Justification for choosing the Activity:

1. **Zero Minute Speech:** The understanding level of the students about the topic will be analysed.
2. **Collaborative Study:** Discussion between small groups happens and learning occurs within the group.
3. **Minute Paper:** The students' attention inside the class will be observed.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

1. **Zero Minute Speech:** The students can present anytime without the need of prior preparation which enables them to listen to the class effectively. The students involved and deliver the speech.
2. **Collaborative Study:** Enhancing the learning among the students through effective discussions and debate.
3. **Minute Paper:** It provides full attention by the students during class to involve in this activity.

Innovative Teaching Method Execution

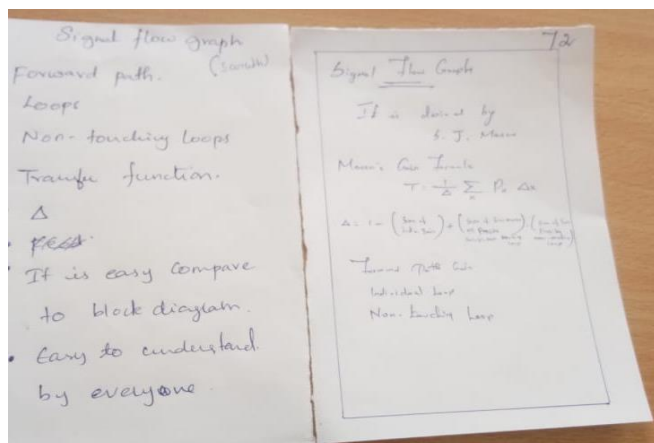
Zero Minute Speech for the Topic Control System Basic Structure on 01.07.2019



Collaborative Study for the Topic Block Diagram Models - Reduction Procedure on 08.07.2019



Minute Paper for the Topic Signal Flow Graphs Models & Block Diagram Reduction on 10.07.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- ❖ **Zero Minute Speech:** The students effectively explain about basic structure of control system.
- ❖ **Collaborative Study:** Students split into groups and all the groups participated well. Learning happens among the group.
- ❖ **Minute Paper:** The students recollect the topics and write effectively.

Changes required for future (if required):

- More students may be given a chance for zero minute speech.
- Time duration need to be increased for collaborative Study.

Reflective Critique:**Challenges:**

1. **Zero Minute Speech:** Some students are not ready to deliver speech.
2. **Collaborative Study:** Time is not enough for some groups to complete the task..
3. **Minute Paper:** All the students could not recollect and write the topics.

Benefits:

1. **Zero Minute Speech:** Students actively involved in this activity and share their views about the topic.
2. **Collaborative Study:** Enhances the students for better learning through healthy discussion.
3. **Minute Paper:** Creates students attention during class hours and they can write the topic effectively.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	1	-	-	1	1	1	1	1

CO1: The students will be able to perform modeling of control system using various techniques.

References:

- A.Nagoor Kani, "Control Systems Engineering", RBA Publications, First Edition, 2013.
- <http://archive.wceruw.org/cl1/CL/moreinfo/MI2A.htm>
- <https://k12teacherstaffdevelopment.com/tlb/how-can-i-use-the-minute-paper-strategy-to-enhance-learning-in-the-classroom/>
- <https://www.teacherswithapps.com/1-minute-speech-a-universal-tool-for-all-teachers/>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-II

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Dr.B.Deepa Lakshmi

Name of the Topic: Time Response of First Order and Second Order Systems & PD, PI, PID control systems

Name of the Innovative Practice: Simulation Tool & Mind Map

Date & Duration: 29.07.2019 (20 Minutes) & 06.08.2019 (20 Minutes)

Description

(i) Simulation Tool:

- Simulation tool is used to correlate the theoretical concepts using MATLAB. It enables the students to observe the working of the given system practically.
- The advantage of using the simulation tool is that it enhances the knowledge of the students by doing it practically. It also determines the accuracy of the system when compared with theoretical calculations.

Goal (Learning Outcomes):

The students will be able to obtain the time response of first order and second order system using this MATLAB Software simulation tool.

Use of appropriate method:

Justification for choosing the Activity:

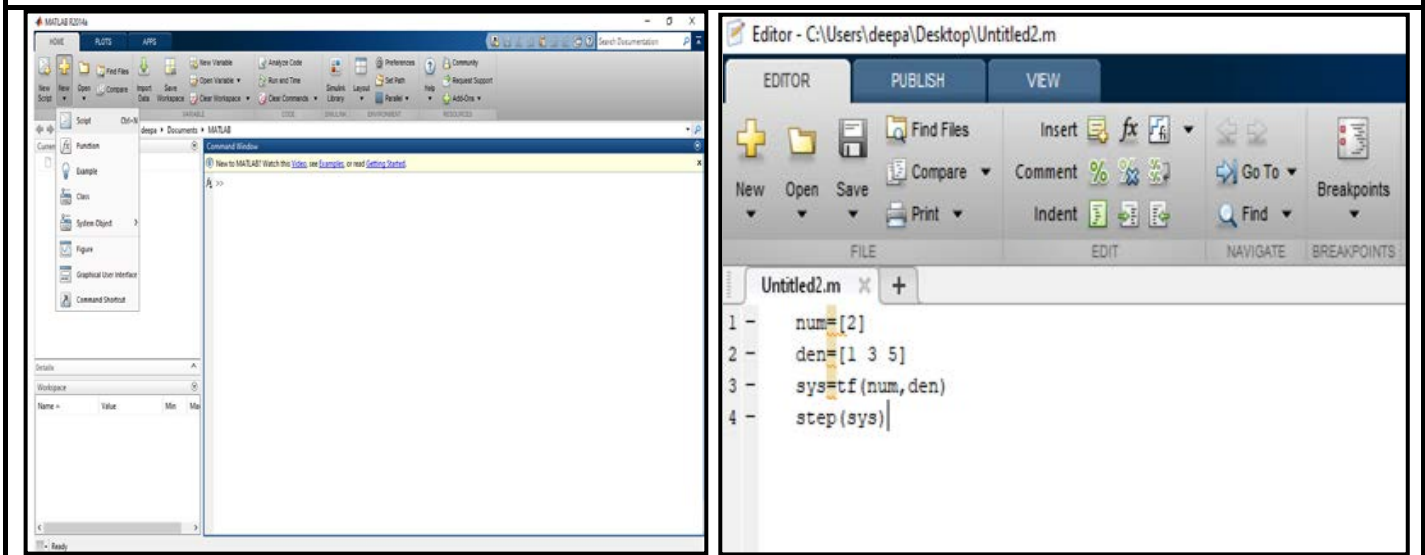
It enhances the technical knowledge of the students through practical exposure and better understanding of the topic.

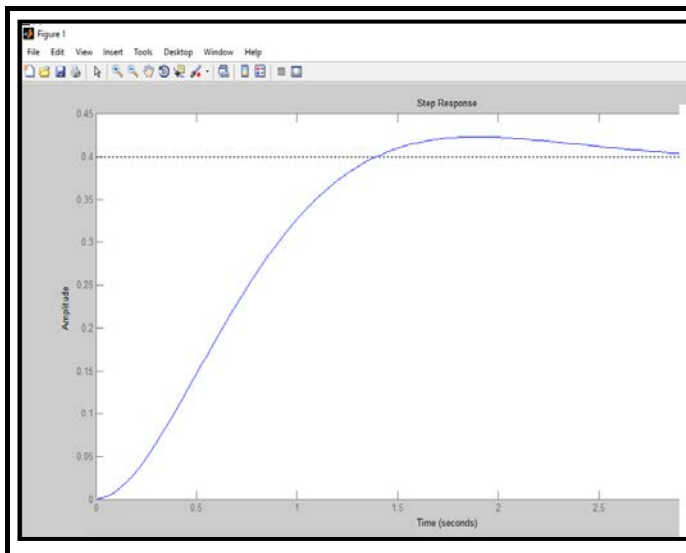
Effective Presentation: (Implementation (Plan & Execution) with Proof):

- It takes more time to explain the response of first and second order system.
- Basic knowledge of MATLAB software needed for effective implementation.

Innovative Teaching Method Execution

Simulation Tool





Significance of Results: (Assessment of Effectiveness/Success of the Activity):

- Using MATLAB software simulation tool, the students could able to analyse the response of any system.
- It provides lifelong learning for the students.

Changes required for future (if required):

- The students should know the basics in using MATLAB because this activity takes more time.
- The students should be informed earlier about this activity for effective implementation.

Reflective Critique:

Challenges:

The students face difficulties in following the procedure, so it takes more time to explain the concepts.

Benefits:

Using MATLAB software simulation tool, the students could be able to analyse the response of first order and second order system.

Description:

(ii) Mind Map:

- A mind map is an easy way to analyse the understanding of the students after completing the topic. It enhances the thinking ability and the students can visualize the content to recall.
- A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Goals (Learning Outcomes):

The students will be able to compare and explain the P, PI and PID controllers.

Use of appropriate method:

Justification for choosing the Activity:

It helps the students for visualizing the ideas given. The mind map enables the students to explain about their own view.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

Students were not able to follow the task hence explained about the concepts and activity again.

Innovative Teaching Method Execution**Mind Map**

Proportional controller:

$$u(t) \propto e(t)$$

$$u(t) = k_p e(t)$$

$$U(s) = k_p E(s)$$

Integral controller:

$$u(t) \propto \int e(t) dt$$

$$u(t) = k_i \int e(t) dt$$

$$U(s) = k_i \frac{E(s)}{s}$$

Proportional and Integral controller:

$$u(t) \propto [e(t) + \int e(t) dt]$$

$$u(t) = k_p e(t) + \frac{k_p}{T_i} \int e(t) dt$$

$$U(s) = k_p \left[1 + \frac{1}{T_i s} \right] E(s)$$

Proportional Integral & Derivative controller:

$$u(t) \propto [e(t) + \int e(t) dt + \frac{d}{dt} e(t)]$$

$$u(t) = k_p e(t) + \frac{k_p}{T_i} \int e(t) dt + k_p T_d \frac{d}{dt} e(t)$$

$$U(s) = k_p \left[1 + \frac{1}{T_i s} + s T_d \right] E(s)$$

Significance of Results: (Assessment of Effectiveness/Success of the Activity):

Due to this activity, Brainstorming happens with the students and they could explain their views.

Changes required for future (if required):

The students should be informed earlier about the topic for better implementation of the activity..

Reflective Critique:**Challenges:**

Students normally hesitate to interact and this enables the students to explain about the topic.

Benefits:

- Brainstorming happens with the students and they could explain their views.
- Used to correct the concepts they understood while teaching in a better way.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	3	3	1	1	1	-	-	1	1	1	1	1

CO2: The students will be able to obtain the time response and steady state error of control systems.

References:

- A.Nagoor Kani, “Control Systems Engineering”, RBA Publications, First Edition, 2013.
- <https://in.mathworks.com/help/control/examples/plotting-system-responses.html>
- <http://www.inspiration.com/visual-learning/mind-mapping>
- <https://www.sdera.wa.edu.au>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-III

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Dr.B.Deepa Lakshmi

Name of the Topic: Bode Plot and Polar Plot

Name of the Innovative Practice: Audio & Video Tools & Buzz Session

Date & Duration: 17.08.2019 (20 Minutes) & 21.08.2019 (20 Minutes)

Description:

(i) Audio & Video Tools:

- ❖ Audio-visual tool is used to improve the teaching learning process. It provides effective communication between teacher and student for understanding the concept.
- ❖ It enables the students to remember the concepts for longer period of time. Students' attention is happened while using this tool.
- ❖ Eventhough chalk n talk method made the students to understand the concept but audio-video tool enhances better understanding by the students in an easy way.

Goals (Learning Outcomes):

The students will be able to obtain the phase margin and gain margin for the given system using Bode plot and Polar Plot.

Use of appropriate method:

Justification for choosing this activity:

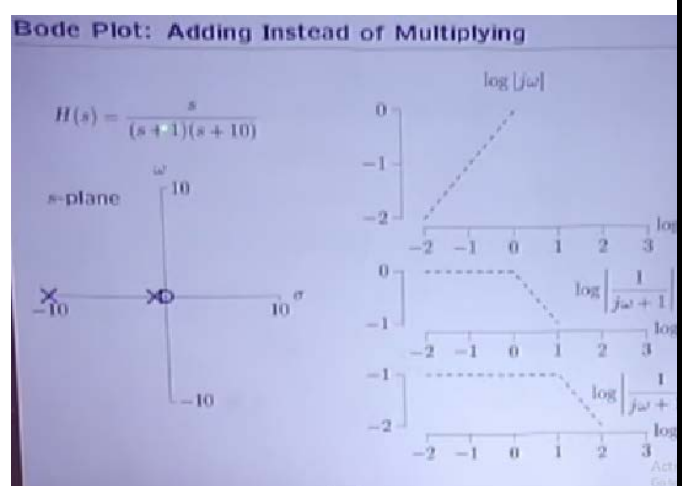
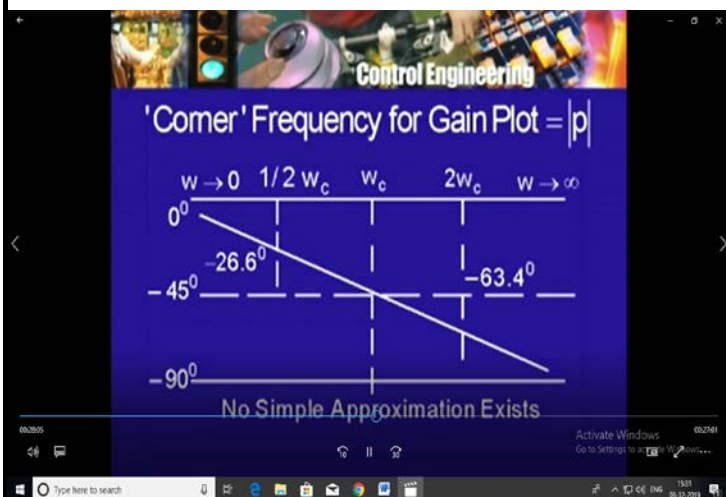
This activity enhances better understanding of the topic which will be remembered by the students for a longer period of time and could able to explain by the students at any time.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

Since video is running for more time, highlight of the important video clips were shown to the students.

Innovative Teaching Method Execution

Audio & Video Tools for the Topic Bode plot on 17.08.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

By visualizing the concepts through this activity, the students could be able to understand the concepts in a better way.

Changes required for future (if required):

Full Video may be shown to the students or video with less time may be selected for this activity.

Reflective Critique:**Challenges:**

This activity takes more time and some students are not concentrating in listening the videos.

Benefits:

The students gain more knowledge while seeing the process visually and could be able to explain the topic anytime. This enhances the student learning.

Description:**(ii) Buzz Session:**

- ❖ This session involves a small group usually involving 3 persons to answer for the questions. It activates the discussions between the students and made them to debate for the correct answers.
- ❖ The whole class is divided into small groups. The group spend very less period of time and answer the question. Each group then report and again a common discussion will take place in order to share their findings and discuss the concepts.

Goals (Learning Outcomes):

The students will be able to obtain the phase margin and gain margin for the system.

Use of appropriate method:**Justification for choosing this activity:**

Buzz sessions are short participating sessions that are used to discuss about their ideas in groups and also in common to explain their understanding.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

The students were informed about the activity and then splitted into groups. They were actively participated for this activity.



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

Since the group is small, they could discuss effectively to bring out answers for the questions.

Changes required for future (if required):

Students need more time to discuss, so time may be extended in future.

Reflective Critique:**Challenges:**

There is no effective participation by the individuals. Some students involve and some are not showing interest and involvement.

Benefits:

The students will start discuss about the given topic immediately. More ideas will be disseminated among their groups. It increases effective communication between the peers and learning happens.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	3	1	1	1	-	-	1	1	1	1	1

CO3: The students will be able to design various compensators and to analyse the frequency response of the system using various plots.

References:

- ✓ A.Nagoor Kani, "Control Systems Engineering", RBA Publications, First Edition, 2013.
- ✓ <https://elearningindustry.com/the-8-best-free-video-tools-for-teachers>
- ✓ <http://sabarishedn.blogspot.com/2015/06/techniques-of-teaching-science-brain.html>
- ✓ <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-003-signals-and-systems-fall-2011/lecture-videos/lecture-11-continuous-time-ct-frequency-response-and-bode-plot/>

RAMCO INSTITUTE OF TECHNOLOGY
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Innovative Practices Description for Unit-IV

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Dr.B.Deepa Lakshmi

Name of the Topic: Routh stability criterion & Root locus

Name of the Innovative Practice: Zero Minute Speech & Audio & Video Tools (MIT Open Courseware)

Date & Duration: 14.09.2019 (5 Minutes) & 19.09.2019 (15 Minutes)

Description:

(i) Zero Minute Speech:

Zero Minute Speech enables the students to explain the topic in short in a better way. This can be understood by other students easily. So, the students will listen the speech and they will be ready to give speech on the topic.

Goals (Learning Outcomes):

The students will be able to obtain the stability of the control system.

Use of appropriate method:

Justification for choosing the Activity:

This activity enables the student to express the topic through his understanding level which can be disseminated to other students.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

The students could immediately deliver the speech without prior preparation for this activity.



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

The students could explain about the topics and clarify their doubts among them in this activity.

Changes required for future (if required):

Time may be increased for this activity to conduct in a better way.

Reflective Critique:**Challenges:**

While asking the students for this activity, some students are not volunteered to give speech. Remaining students in the class are interested to deliver.

Benefits:

The topic shared by any student will be easily grasped by the student friends and they clarify doubts with the friends.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	1	1	1	-	-	1	1	1	1	1

CO4: The students will be able to determine the stability of control systems.

Description:**(ii) Audio & Video Tools:**

- Audio-visual tool is used to improve the teaching learning. It provides effective communication between teacher and students for understanding the concept.
- It enables the students to remember the concepts for longer period of time. Students' attention is happened while using this tool.
- Even though chalk n talk method made the students to understand but audio video tool enhances better understanding by the students in an easy way.

Goals (Learning Outcomes):

The students will be able to plot the root locus of the system and determine stability of the system.

Use of appropriate method:**Justification for choosing the Activity:**

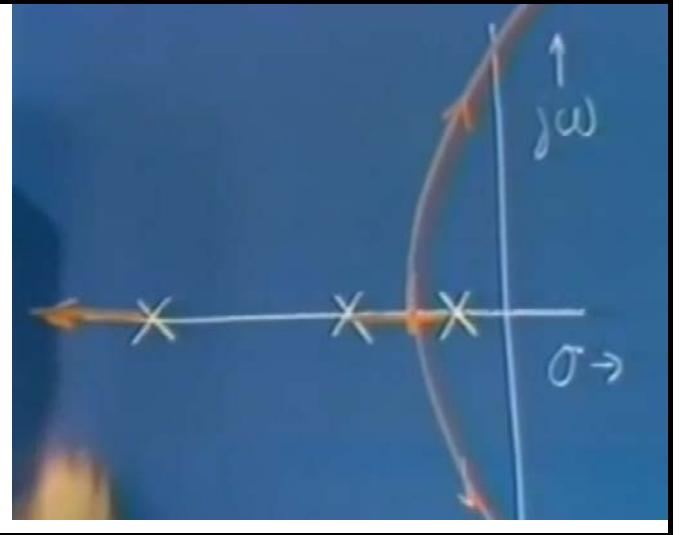
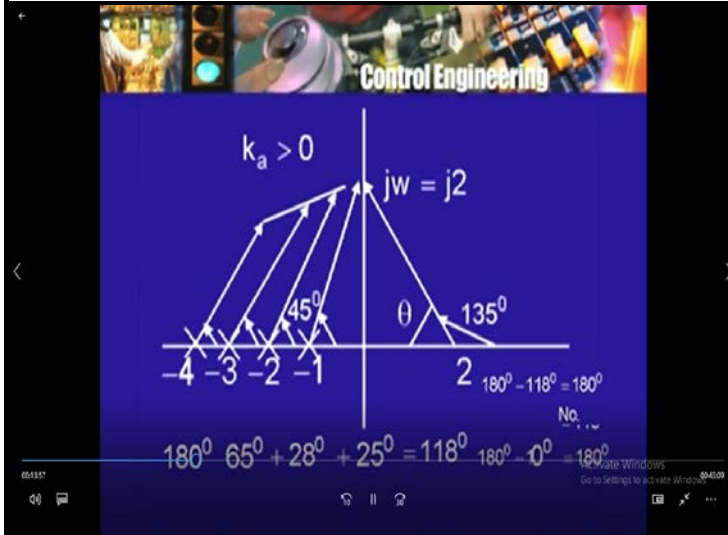
This activity enhances better understanding of the topic which will be remembered by the students for a longer period of time and could able to explain by the students at any time.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

Highlights of the video were shown to the students for better understanding. It reduces the time for this activity.

Innovative Teaching Method Execution

Audio & Video Tools for the Topic Root locus on 19.09.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

Students could be able to understand the concepts clearly while visualizing it. This enhances better learning.

Changes required for future (if required):

Videos with less time to be chosen for this activity or one full hour to be utilized for displaying the video.

Reflective Critique:

Challenges:

This activity takes more time and some students are not concentrating in listening the videos.

Benefits:

The students gain more knowledge while seeing the process visually and could be able to explain the topic anytime. This enhances the student learning.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	3	2	1	1	1	-	-	1	1	1	1	1

CO4: The students will be able to determine the stability of control systems.

References:

- A.Nagoor Kani, "Control Systems Engineering", RBA Publications, First Edition, 2013.
- <https://www.teacherswithapps.com/1-minute-speech-a-universal-tool-for-all-teachers>
- <https://www.youtube.com/watch?v=KLylS0Tbr20>

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019-2020 (Odd Semester)

Innovative Practices Description for Unit-V

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8391 Control Systems Engineering

Name of the Faculty member: Dr.B.Deepa Lakshmi

Name of the Topic: Controllability and Observability & Digital control design using state feedback

Name of the Innovative Practice: Exit Slips & Brain storming

Date & Duration: 30.09.2019 (10 Minutes) & 09.10.2019 (15 Minutes)

Description:

(i) Exit Slips:

It enables the teacher to measure the level of understanding the topic by the students. It enhances the thinking ability of the students. The student will remember the concept once they write it immediately and remember for a longer period of time.

Goals (Learning Outcomes):

The students will be able to recollect the concept and determine controllability and observability.

Use of appropriate method:

Justification for choosing the Activity:

It allows the faculty member to measure the understanding level of the students. Interest will be created among the students

Effective Presentation: (Implementation (Plan & Execution) with Proof):

The understanding level of the students may be measured in this activity. The students could able to recollect the concepts taught in the class.

Innovative Teaching Method Execution

Exit Slips for the Topic Controllability and Observability on 30.09.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

The students' involvement is better which leads to recollect the topics and write it which enhances better learning.

Changes required for future (if required):

The time duration may be extended for the students to write about the topic in this activity.

Reflective Critique:**Challenges:**

All the students could not remember the concept and write immediately. SO, some students were unable to write the formulae.

Benefits:

The Students remembering level will increase while conducting this activity. It also enhances teaching learning.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	2	1	1	1	-	-	1	1	1	1	1

Description:**(ii) Brain storming:**

Brainstorming is a small group activity which enables to throw their view about a particular topic. It generates more ideas among the students. It provides opportunity to communicate with the peers without any hesitation.

Goals (Learning Outcomes):

The students will be able to analyse the concept while sharing their views.

Use of appropriate method:**Justification for choosing the Activity:**

This creates more opportunities for the students to share their views.

Effective Presentation: (Implementation (Plan & Execution) with Proof):

The understanding level of the students could be analysed in this activity. The students recollect and present their views about the topic.

Innovative Teaching Method Execution

Brain storming for the Topic Digital control design using state feedback on 09.10.2019



Significance of Results: (Assessment of Effectiveness/Success of the Activity):

The students explained the topics in a better way after discussion among them. It benefits the students in understanding the topics.

Changes required for future (if required):

The time duration need to be increased for this activity in future which leads to discuss more about the topic by the students.

Reflective Critique:**Challenges:**

It requires more time while discuss in groups. So, time limit to be increased for this activity.

Benefits:

The student members among the group will start sharing their views while sitting in small groups. Sharing of ideas by the individual makes other students to learn and understand.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	3	2	1	1	1	-	-	1	1	1	1	1

CO5: The students will be able to analyse and obtain state space model using state variables.

References:

- https://www.readingrockets.org/strategies/exit_slips
- <https://teaching.unsw.edu.au/brainstorming>
- <https://www.youtube.com/watch?v=GLpZ6RZHyoM>